



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

Academic Year (2024-2025)

Program: B. Tech in CSE(ICB)

Year: T. Y B. Tech

Semester: VI

Theory Course: Data Engineering Laboratory (DJS22ICL602)

List of Laboratory Experiments: (Minimum any eight experiments)	
Sr. No.	Suggested Experiments
1	To Implement Loading the data from CSV or JSON files using SQL command (PostgreSQL) or LOAD DATA (MySQL).
2	To Build an end-to-end ETL pipeline: extract data from an API, transform it and load it into a database.
3	To Load a large dataset into a data warehouse and run queries on it. (e.g., Google or BigQuery).
4	To Implement SQL-based transformations using data build tool to build a reliable data transformation pipeline.
5	To Build a data pipeline that applies multiple transformations to a dataset, creating a repeatable process
6	Implement least privilege principle to minimize the number of permissions granted to users by using database tool.
7	To implement Data Anonymization masking and redaction techniques to protect sensitive information.
8	To Test different data quality metrics and validation techniques by using Apache Spark.
9	To Share sensitive data with external organizations using Lake Formation's data sharing capabilities.
10	To perform fine-grained access control policies on tables and columns within your data lake.
11	To Configure Kinesis Data Firehose for streaming delivery to Amazon S3.
12	To Deploy dataset as an Infrastructure Using Apache Spark.
13	To Create Delta Lake Table and demonstrate how Delta Lake Enables the Lake House.
14	To implement data cataloguing and Sharing Services by using open-source tools Apache Atlas.
15	To Create Pipeline for the Gold Layer by using ETL Tools (Apache Spark or dbt).

Mrs. Shilpa M. Satre
Subject In charge

Dr. Narendra Shekokar
Head of the Department



**SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING**
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



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